THE APERIODIC JORDAN ALGEBRAS OF QUASICRYSTALS ARE NOT JORDAN

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Abstract. Corradetti, Chester, Aschheim and Irwin attached to every quasicrystal closed under quasiaddition an algebra with product $L_x \circ L_y = \frac{1}{2}(L_{x\vdash y} + L_{y\vdash x})$ and stated that it is a Jordan algebra. It is not even power-associative: $x^2x^2 = x^3x$ holds exactly when x is a multiple of a generator.

1. Introduction

Quasicrystal Lie algebras were introduced by Patera, Pelantová and Twarock [10]: their generators L_x are indexed by the points x of a one-dimensional quasicrystal, and the bracket $[L_x, L_y]$ is a multiple of L_{x+y} when $x + y$ lies again in the quasicrystal and vanishes otherwise. Corradetti, Chester, Aschheim and Irwin replaced the sum by the *quasiaddition* of Berman and Moody [3],

$$x \vdash y = \tau^2 x - \tau y, \qquad \tau = \tfrac{1+\sqrt{5}}{2}, \tag{1}$$

under which every model set with a convex window is closed, and defined on the span of the generators the commutative product

$$L_x \circ L_y = \tfrac{1}{2}\left(L_{x\vdash y} + L_{y\vdash x}\right). \tag{2}$$

In [7, Section 5] this is done for the Fibonacci-chain quasicrystals and the resulting algebra is called "the first aperiodic Jordan algebra"; in [8, Section 5] the same product is placed on every icosahedral model set with a convex window, in dimensions one to four, and a theorem states that the algebra is a Jordan algebra. Both papers go on to derive structural properties from the Jordan property, and [7] proposes, through Svinolupov's correspondence between Jordan algebras and integrable systems of Korteweg–de Vries type [11], an application of finite truncations of the algebra to solitons.[1]

The proof given in both papers checks the Jordan identity

$$(x \circ y) \circ (x \circ x) = x \circ \big(y \circ (x \circ x)\big) \tag{3}$$

for $x = L_p$ and $y = L_q$, using $p \vdash p = p$. Since (3) is cubic in x, its validity on a basis says nothing about sums of basis vectors; a check on generators would have to be made on the full linearisation of the identity [9, Part II]. We show that the identity fails for every element outside the lines spanned by the generators. Throughout, Ξ is a subset of a real vector space V that is closed under (1) and has at least two points, and $\mathfrak{J}(\Xi)$ is the real vector space with basis $\{L_x\}_{x\in\Xi}$ and product (2). We write $x^2 = x \circ x$ and $x^3 = x^2 \circ x$.

Theorem A. *For $x \in \mathfrak{J}(\Xi)$ the following are equivalent.*

(i) *$x = cL_p$ for some $c \in \mathbb{R}$ and $p \in \Xi$.*

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[1] We quote the arXiv versions arXiv:2302.04044 and arXiv:2303.12219 of [7, 8]. In [7] the claim is the construction of Section 5 together with its Conclusions; in [8] it is the first theorem of Section 5.

(ii) *The Jordan identity* (3) *holds for* x *and every* $y \in \mathfrak{J}(\Xi)$.
(iii) $x^2 \circ x^2 = x^3 \circ x$.

In particular $\mathfrak{J}(\Xi)$ *is neither a Jordan algebra nor a power-associative algebra.*

A Jordan algebra is power-associative [1, 9], and in a commutative algebra power-associativity in degree four is exactly (iii); so Theorem A shows that $\mathfrak{J}(\Xi)$ already fails the weakest consequence of the Jordan identity, and that the computation in [7, 8], which is the implication (i)⇒(ii), is the whole range in which the identity holds. The cases covered include every Fibonacci-chain quasicrystal $\mathcal{F}_{\alpha,\beta}$ of [7] and the Fibonacci chain, the Penrose tiling and the Elser–Sloane quasicrystal of [8]. For $x = L_1 + L_{1+\tau}$ in the Fibonacci chain $\mathcal{F}_{1,0}$ the generator $L_{9+13\tau}$ occurs in $x^3 \circ x$ with coefficient $\frac{1}{4}$ and does not occur in $x^2 \circ x^2$ (Example 4.2 and Figure 1).

The finite algebras proposed in [7, Section 6] are obtained by keeping finitely many consecutive points of the chain and setting a product to zero when a quasiaddition leaves them; [7] states that commutativity and the Jordan identity survive this truncation.

Theorem B. *Truncation does not produce Jordan algebras. Keeping the three consecutive points* $1,\ 1+\tau,\ 2+2\tau$ *of* $\mathcal{F}_{1,0}$ *gives the three-dimensional commutative algebra with basis* e_1, e_2, e_3 *and products* $e_i^2 = e_i$, $e_1e_2 = \frac{1}{2}e_3$, $e_1e_3 = e_2e_3 = 0$, *in which* $x = e_1 + e_2$ *satisfies* $x^2x^2 - x^3x = e_3$. *More generally, for every* $x \in \mathfrak{J}(\Xi)$ *with at least two points in its support there is a finite set* $T \subset \Xi$ *such that every truncation to a finite set* $S \supseteq T$ *fails power-associativity.*

Finally we examine the structural statements that [7, 8] derive for these algebras.

Theorem C.
(a) *The subspace* $\mathfrak{k} = L_1 \circ \mathfrak{J}(\mathcal{F}_{1,0})$, *which* [7] *uses as a proper ideal to show that* $\mathfrak{J}(\mathcal{F}_{1,0})$ *is not simple, is not an ideal:* $(L_1 \circ L_{1+\tau}) \circ L_{2+3\tau} \notin \mathfrak{k}$.
(b) *Nevertheless* $\mathfrak{J}(\Xi)$ *is not simple: the coefficient sum is a nonzero algebra homomorphism* $\mathfrak{J}(\Xi) \to \mathbb{R}$.
(c) $\mathfrak{J}(\Xi)$ *has no unit, its one-dimensional subalgebras are the lines* $\mathbb{R}L_x$, *and it has no finite-dimensional subalgebra of dimension at least two.*
(d) *No algebra containing* $\mathfrak{J}(\Xi)$ *as a subalgebra is a Jordan algebra; in particular the unital extension of* $\mathfrak{J}(\Xi)$ *described in* [8, Section 5] *is not one.*

Parts (b) and (c) confirm conclusions of [7, 8], but the proofs given there do not establish them: the argument for non-simplicity in [7] rests on the subspace in (a), the arguments for the absence of a unit in both papers manipulate the infinite sum $\sum_{x\in\Xi} L_x$, which is not an element of the algebra, and the argument for the absence of finite-dimensional subalgebras in [8] assumes that such a subalgebra contains two generators L_x, L_y.

The proofs rest on one observation. Quasiaddition commutes with affine maps, so after choosing a linear functional that separates the points of a support, each product of finitely supported elements has a highest and a lowest point whose coefficients can be read off exactly (Lemmas 3.1 and 3.2). Following the highest point through x^2, x^3, $x^2 \circ x^2$ and $x^3 \circ x$ proves Theorem A. Other questions raised by the construction of Rios, Marrani and Chester in a different setting are treated in [5, 2], and further algebraic invariants attached to geometric data are studied in [4, 6].

2. Quasiaddition and the algebra $\mathfrak{J}(\Xi)$

Let V be a real vector space and let $\tau = \frac{1+\sqrt{5}}{2}$, so that $\tau^2 = \tau + 1$. Repeated use of this relation gives

$$\tau^3 = 2\tau + 1, \quad \tau^4 = 3\tau + 2, \quad \tau^5 = 5\tau + 3, \quad \tau^6 = 8\tau + 5, \quad \tau^7 = 13\tau + 8. \tag{4}$$

Quasiaddition (1) can be written as $x \vdash y = y + \tau^2(x - y)$, because $\tau^2 - \tau = 1$.

Lemma 2.1. *Let $x, y \in V$.*

(a) $x \vdash x = x$.
(b) *If $\varphi\colon V \to W$ is an affine map into a real vector space W, then $\varphi(x \vdash y) = \varphi(x) \vdash \varphi(y)$. In particular $f(x \vdash y) = \tau^2 f(x) - \tau f(y)$ for every affine function $f\colon V \to \mathbb{R}$.*
(c) $x \vdash y = x$ *if and only if* $y = x$, *and* $y \vdash x = x$ *if and only if* $y = x$.
(d) *If $\Xi \subset V$ is closed under quasiaddition and contains two points $a \neq b$, then Ξ is infinite.*

Proof. (a) $x \vdash x = (\tau^2 - \tau)x = x$. (b) Write $\varphi(v) = Av + w$ with A linear. Then $\varphi(x \vdash y) = \tau^2 Ax - \tau Ay + w = \tau^2(Ax + w) - \tau(Ay + w)$, since $\tau^2 - \tau = 1$; this is $\varphi(x) \vdash \varphi(y)$. (c) $x \vdash y = x$ means $(\tau^2 - 1)x = \tau y$, that is $\tau x = \tau y$; and $y \vdash x = x$ means $\tau^2 y = (1 + \tau)x = \tau^2 x$. (d) Put $z_0 = b$ and $z_{k+1} = z_k \vdash a$. By the second form of (1), $z_{k+1} - a = \tau^2(z_k - a)$, so $z_k - a = \tau^{2k}(b - a)$, and these vectors are pairwise distinct because $\tau^2 > 1$ and $b \neq a$. All z_k lie in Ξ. □

For a subset $\Xi \subset V$ closed under quasiaddition we write $\mathfrak{J}(\Xi)$ for the real vector space with basis $\{L_x\}_{x \in \Xi}$ and the bilinear product (2). An element is written $u = \sum_x a_x L_x$ with finitely many nonzero a_x; its *support* $\operatorname{supp} u$ is the set of x with $a_x \neq 0$. Expanding (2) bilinearly, for $u = \sum_x a_x L_x$ and $v = \sum_y b_y L_y$,

$$u \circ v = \tfrac{1}{2} \sum_{x,y} a_x b_y \left(L_{x \vdash y} + L_{y \vdash x} \right), \tag{5}$$

the sum running over $x \in \operatorname{supp} u$ and $y \in \operatorname{supp} v$. When $v = u$ the two halves of (5) are exchanged by renaming $x \leftrightarrow y$, hence

$$u \circ u = \sum_{x,y \in \operatorname{supp} u} a_x a_y L_{x \vdash y}. \tag{6}$$

In particular $L_x \circ L_x = L_x$ by Lemma 2.1(a), and the product is commutative.

The quasicrystals of [7, 8] are of this kind. A Fibonacci-chain quasicrystal is $\mathcal{F}(\Omega) = \{n \in \mathbb{Z}[\tau] : n^* \in \Omega\} \subset \mathbb{R}$, where Ω is an interval of length one and $(n_1 + n_2\tau)^* = n_1 + n_2(1 - \tau)$ is the Galois conjugation; the sets $\mathcal{F}_{\alpha,\beta}$ of [7] are the cases $\Omega = (\alpha + \beta - 1, \alpha + \beta]$. Since conjugation is a ring automorphism of $\mathbb{Z}[\tau]$ and $(1 - \tau)^2 - (1 - \tau) = 1$, it satisfies $(x \vdash y)^* = (1 - \tau)^2 x^* - (1 - \tau) y^* = y^* + (1 - \tau)^2 (x^* - y^*)$, and this point lies between x^* and y^* because $0 < (1 - \tau)^2 < 1$; so a convex window gives a set closed under (1). The icosahedral model sets of [8] in $\mathbb{R}^2$, $\mathbb{R}^3$ and $\mathbb{R}^4$ are closed under (1) for the same reason [8, Section 4]. We shall use only closure under (1).

For $\mathcal{F}_{1,0}$, where $\Omega = (0, 1]$, the points near the origin are

$$\ldots,\ -1 - 3\tau,\ -1 - 2\tau,\ -\tau,\ 1,\ 1 + \tau,\ 2 + 2\tau,\ 2 + 3\tau,\ 3 + 4\tau,\ \ldots$$

For instance $(2 + 3\tau)^* = 5 - 3\tau \in (0, 1]$, while $0^* = 0$ and $\tau^* = 1 - \tau$ are not in $(0, 1]$, so $0, \tau \notin \mathcal{F}_{1,0}$.

3. The leading terms of a product

Fix an affine function $f\colon V \to \mathbb{R}$. For a nonzero $u = \sum a_x L_x$ put

$$M_f(u) = \max_{x \in \operatorname{supp} u} f(x), \qquad m_f(u) = \min_{x \in \operatorname{supp} u} f(x).$$

We call u *f-sharp* if exactly one point of $\operatorname{supp} u$, denoted $u^{\uparrow}$, has $f(u^{\uparrow}) = M_f(u)$, and exactly one point, denoted $u^{\downarrow}$, has $f(u^{\downarrow}) = m_f(u)$. Their coefficients $u_{\uparrow} = a_{u^{\uparrow}}$ and $u_{\downarrow} = a_{u^{\downarrow}}$ are then nonzero.

Lemma 3.1. *Let u be f-sharp with $M = M_f(u) > m = m_f(u)$. Then $u \circ u$ is f-sharp, with*

$$M_f(u \circ u) = \tau^2 M - \tau m, \qquad m_f(u \circ u) = \tau^2 m - \tau M,$$

$(u \circ u)^{\uparrow} = u^{\uparrow} \vdash u^{\downarrow}$ and $(u \circ u)^{\downarrow} = u^{\downarrow} \vdash u^{\uparrow}$, and both of these points carry the coefficient $u_{\uparrow}u_{\downarrow}$. Moreover $M_f(u \circ u) - M > 0$ and $M_f(u \circ u) - m_f(u \circ u) = \tau^3(M - m) > 0$.

Proof. By (6) the support of $u \circ u$ is contained in $\{x \vdash y : x, y \in \operatorname{supp} u\}$, and by Lemma 2.1(b)

$$f(x \vdash y) = \tau^2 f(x) - \tau f(y) \le \tau^2 M - \tau m,$$

with equality if and only if $f(x) = M$ and $f(y) = m$, that is, by sharpness, if and only if $(x, y) = (u^{\uparrow}, u^{\downarrow})$. Put $z_0 = u^{\uparrow} \vdash u^{\downarrow}$. The coefficient of L_{z_0} in (6) is the sum of $a_x a_y$ over the pairs with $x \vdash y = z_0$; each such pair has $f(x \vdash y) = f(z_0) = \tau^2 M - \tau m$ and is therefore $(u^{\uparrow}, u^{\downarrow})$. So the coefficient is $u_{\uparrow}u_{\downarrow} \neq 0$, the point z_0 belongs to the support, and every other point z of the support satisfies $f(z) < \tau^2 M - \tau m$, since a point with equality would be $u^{\uparrow} \vdash u^{\downarrow} = z_0$. The statement about the lowest point follows in the same way from $f(x \vdash y) \ge \tau^2 m - \tau M$, with equality exactly for $(x, y) = (u^{\downarrow}, u^{\uparrow})$. Finally $(\tau^2 M - \tau m) - M = \tau(M - m) > 0$ because $\tau^2 - 1 = \tau$, and $(\tau^2 M - \tau m) - (\tau^2 m - \tau M) = (\tau^2 + \tau)(M - m) = \tau^3(M - m)$. □

Lemma 3.2. *Let u and v be f-sharp, and put*

$$A = \tau^2 M_f(u) - \tau m_f(v), \qquad B = \tau^2 M_f(v) - \tau m_f(u),$$
$$a = \tau^2 m_f(u) - \tau M_f(v), \qquad b = \tau^2 m_f(v) - \tau M_f(u).$$

Every point z of $\operatorname{supp}(u \circ v)$ satisfies $\min(a, b) \le f(z) \le \max(A, B)$. If $A > B$, then $u^{\uparrow} \vdash v^{\downarrow}$ is the only point of $\operatorname{supp}(u \circ v)$ with $f = M_f(u \circ v)$, this maximum equals A, and the coefficient of $L_{u^{\uparrow} \vdash v^{\downarrow}}$ in $u \circ v$ is $\frac{1}{2}u_{\uparrow}v_{\downarrow}$. If $a < b$, then $u^{\downarrow} \vdash v^{\uparrow}$ is the only point with $f = m_f(u \circ v)$, this minimum equals a, and its coefficient is $\frac{1}{2}u_{\downarrow}v_{\uparrow}$. In particular, if $A > B$ and $a < b$, then $u \circ v$ is f-sharp.

Proof. By (5) the coefficient of L_z in $u \circ v$ is

$$\tfrac{1}{2} \sum_{x \vdash y = z} a_x b_y + \tfrac{1}{2} \sum_{y \vdash x = z} a_x b_y, \tag{7}$$

with $x \in \operatorname{supp} u$ and $y \in \operatorname{supp} v$ in both sums. A point of the first kind, $z = x \vdash y$, satisfies $f(z) = \tau^2 f(x) - \tau f(y) \le A$, with equality only for $(x, y) = (u^{\uparrow}, v^{\downarrow})$; a point of the second kind, $z = y \vdash x$, satisfies $f(z) = \tau^2 f(y) - \tau f(x) \le B$, with equality only for $y = v^{\uparrow}$ and $x = u^{\downarrow}$. The lower bounds $f(z) \ge a$ and $f(z) \ge b$ are obtained in the same way, which proves the first assertion. Suppose $A > B$ and put $z_0 = u^{\uparrow} \vdash v^{\downarrow}$, so $f(z_0) = A$. In (7) for $z = z_0$ the second sum is empty, because its terms have $f(z) \le B < A$, and the first sum has the single term $(x, y) = (u^{\uparrow}, v^{\downarrow})$, because every term has $f(x \vdash y) = A$. Hence the coefficient of L_{z_0} is $\frac{1}{2}u_{\uparrow}v_{\downarrow} \neq 0$. Any point z of the support with $f(z) = A$ is of the first kind with $(x, y) = (u^{\uparrow}, v^{\downarrow})$, so $z = z_0$; every other point has $f(z) < A$. The case $a < b$ is symmetric. □

4. Proof of Theorem A

Proof of Theorem A. (i)⇒(ii). If $c = 0$ there is nothing to prove. Otherwise $x \circ x = c^2 L_p$, and both sides of (3) are linear in y, so it suffices to take $y = L_q$. By commutativity, used twice,

$$(L_p \circ L_q) \circ L_p = L_p \circ (L_p \circ L_q) = L_p \circ (L_q \circ L_p),$$

and multiplying by c^3 gives $(x \circ L_q) \circ (x \circ x) = x \circ \big(L_q \circ (x \circ x)\big)$.

(ii)⇒(iii). Put $y = x$ in (3). The left side is $x^2 \circ x^2$. On the right, $x \circ (x \circ x) = x^2 \circ x = x^3$ by commutativity, and then $x \circ x^3 = x^3 \circ x$.

(iii)⇒(i). Suppose that $\operatorname{supp} x$ has at least two points; we show that (iii) fails. The finite set $\operatorname{supp} x$ is separated by some linear functional ℓ on V: for each pair $z \neq z'$ in $\operatorname{supp} x$ the functionals vanishing on $z - z'$ form a proper subspace of the dual space, and a real vector space is not a union of finitely many proper subspaces. Let p and q be the points of $\operatorname{supp} x$ where ℓ is largest and smallest, and put

$$f = \frac{\ell - \ell(q)}{\ell(p) - \ell(q)}.$$

This is an affine function with $f(q) = 0$, $f(p) = 1$ and $0 < f < 1$ at the other points of $\operatorname{supp} x$. Thus x is f-sharp with $M_f(x) = 1$, $m_f(x) = 0$, $x^{\uparrow} = p$, $x^{\downarrow} = q$, $x_{\uparrow} = c_p$ and $x_{\downarrow} = c_q$, where c_p and c_q are the coefficients of L_p and L_q in x.

Step 1: x^2. By Lemma 3.1, x^2 is f-sharp with $M_f(x^2) = \tau^2$ and $m_f(x^2) = -\tau$, and both extreme coefficients equal $c_p c_q$.

Step 2: $x^3 = x^2 \circ x$. Apply Lemma 3.2 with $u = x^2$ and $v = x$:

$$A = \tau^2 \cdot \tau^2 - \tau \cdot 0 = \tau^4, \qquad B = \tau^2 \cdot 1 - \tau \cdot (-\tau) = 2\tau^2,$$
$$a = \tau^2 \cdot (-\tau) - \tau \cdot 1 = -\tau^3 - \tau, \qquad b = \tau^2 \cdot 0 - \tau \cdot \tau^2 = -\tau^3.$$

By (4), $A - B = (3\tau + 2) - (2\tau + 2) = \tau > 0$ and $b - a = \tau > 0$. Hence x^3 is f-sharp, $M_f(x^3) = \tau^4$ with coefficient $\frac{1}{2}(c_p c_q)c_q$, and $m_f(x^3) = -\tau^3 - \tau$ with coefficient $\frac{1}{2}(c_p c_q)c_p$.

Step 3: $x^3 \circ x$. Apply Lemma 3.2 with $u = x^3$ and $v = x$:

$$A = \tau^2 \cdot \tau^4 - \tau \cdot 0 = \tau^6, \qquad B = \tau^2 \cdot 1 - \tau \cdot (-\tau^3 - \tau) = \tau^4 + 2\tau^2.$$

By (4), $A - B = (8\tau + 5) - (3\tau + 2) - (2\tau + 2) = 3\tau + 1 > 0$. So $x^3 \circ x$ contains the point $z_* = (x^3)^{\uparrow} \vdash q$, with $f(z_*) = \tau^6$, and the coefficient of L_{z_*} is

$$\tfrac{1}{2} \cdot \tfrac{1}{2} c_p c_q^2 \cdot c_q = \tfrac{1}{4} c_p c_q^3 \neq 0.$$

Step 4: $x^2 \circ x^2$. By Lemma 3.1 applied to $u = x^2$, every point z of $\operatorname{supp}(x^2 \circ x^2)$ satisfies

$$f(z) \leq \tau^2 M_f(x^2) - \tau m_f(x^2) = \tau^4 + \tau^2 = 4\tau + 3 < 8\tau + 5 = \tau^6.$$

Hence $z_* \notin \operatorname{supp}(x^2 \circ x^2)$, and the coefficient of L_{z_*} in $x^2 \circ x^2 - x^3 \circ x$ is $-\frac{1}{4} c_p c_q^3 \neq 0$. Thus $x^2 \circ x^2 \neq x^3 \circ x$.

For the last sentence of the theorem take two distinct points $p, q \in \Xi$ and $x = L_p + L_q$. Then (iii) fails, so the Jordan identity fails for the pair (x, x), and x generates a subalgebra that is not associative, so $\mathfrak{J}(\Xi)$ is not power-associative. □

Remark 4.1. The point z_* is $q + \tau^6(p - q)$. Indeed $(x^2)^{\uparrow} = p \vdash q$ and $(x^3)^{\uparrow} = (x^2)^{\uparrow} \vdash q$ by Lemmas 3.1 and 3.2, so $z_* = ((p \vdash q) \vdash q) \vdash q$, and $w \vdash q - q = \tau^2(w - q)$ applied three times gives $z_* - q = \tau^6(p - q)$.

Example 4.2. In $\mathfrak{J}(\mathcal{F}_{1,0})$ take $x = L_1 + L_{1+\tau}$, so that, with ℓ the identity of $\mathbb{R}$, $q = 1$, $p = 1 + \tau$, $c_p = c_q = 1$ and $f(z) = (z - 1)/\tau$. The quasiadditions of the two points are $1 \vdash 1 = 1$, $(1 + \tau) \vdash (1 + \tau) = 1 + \tau$ and

$$1 \vdash (1 + \tau) = \tau^2 - \tau(1 + \tau) = -\tau, \qquad (1 + \tau) \vdash 1 = \tau^2 + \tau^3 - \tau = 2 + 2\tau,$$

so by (6)

$$x^2 = L_{-\tau} + L_1 + L_{1+\tau} + L_{2+2\tau}.$$

The highest point of x^3 is $(2 + 2\tau) \vdash 1 = 2\tau^2 + 2\tau^3 - \tau = 4 + 5\tau$, with coefficient $\frac{1}{2}$, and the highest point of $x^3 \circ x$ is

$$z_* = (4 + 5\tau) \vdash 1 = 4\tau^2 + 5\tau^3 - \tau = 9 + 13\tau = 1 + \tau^7,$$

with coefficient $\frac{1}{4}$, in accordance with Remark 4.1. Every point of $x^2 \circ x^2$ is at most $(2+2\tau) \vdash (-\tau) = 2\tau^2 + 2\tau^3 + \tau^2 = 5 + 7\tau < 9 + 13\tau$. Hence $x^2 \circ x^2 - x^3 \circ x$ contains $-\frac{1}{4} L_{9+13\tau}$. All supports involved lie in $\mathcal{F}_{1,0}$, which is closed under quasiaddition; Figure 1 shows them.

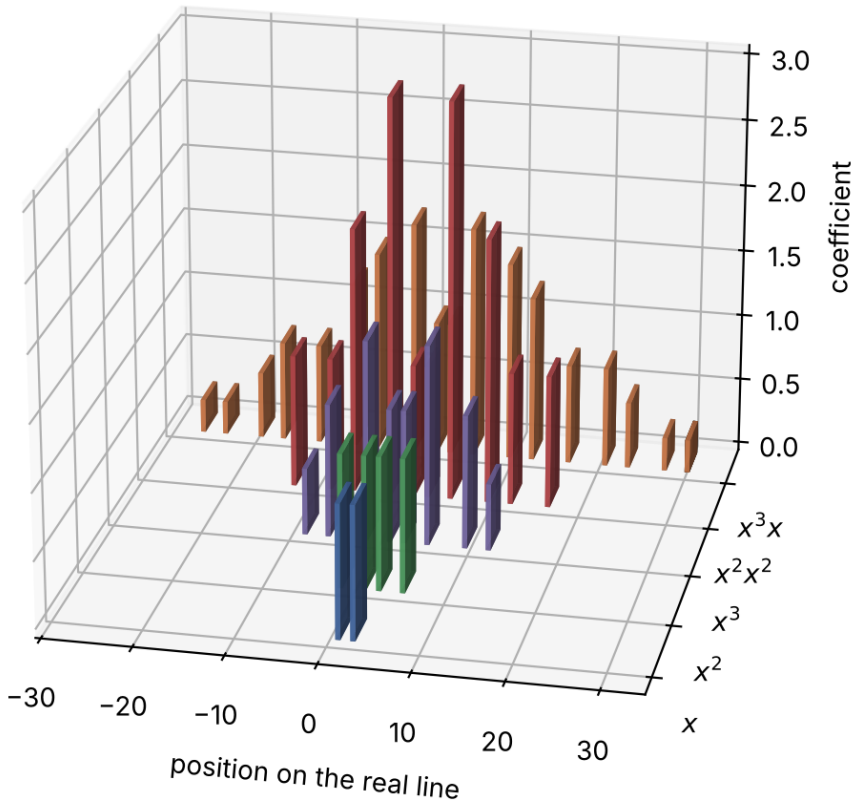


FIGURE 1. Supports and coefficients of x, x^2, x^3, x^2x^2 and x^3x for $x = L_1 + L_{1+\tau}$ in $\mathfrak{J}(\mathcal{F}_{1,0})$, plotted at the positions of the points on the real line. The rightmost bar of x^3x is $L_{9+13\tau} \approx L_{30.03}$ with coefficient $\frac{1}{4}$; the support of x^2x^2 ends at $5 + 7\tau \approx 16.33$.

5. Finite truncations: proof of Theorem B

For a finite subset $S \subset \Xi$ let π_S be the linear map that keeps L_z for $z \in S$ and sends L_z to 0 for $z \notin S$. Let $\mathfrak{J}_S$ be the span of $\{L_x\}_{x \in S}$ with the product

$$L_x \circ_S L_y = \tfrac{1}{2} \left(\pi_S L_{x \vdash y} + \pi_S L_{y \vdash x} \right). \tag{8}$$

This is the truncation described in [7, Section 6], where the product is set to zero whenever a quasiaddition leaves the chosen points. One may also read that description as setting $L_x \circ_S L_y = 0$ as soon as one of $x \vdash y$, $y \vdash x$ lies outside S; the second part of Theorem B holds for both readings.

Proof of Theorem B. The three-point truncation. Let $S = \{1, 1+\tau, 2+2\tau\}$, three consecutive points of $\mathcal{F}_{1,0}$, and put $e_1 = L_1$, $e_2 = L_{1+\tau}$, $e_3 = L_{2+2\tau}$. By Lemma 2.1(a), $e_i \circ_S e_i = e_i$. Example 4.2 gives $1 \vdash (1+\tau) = -\tau \notin S$ and $(1+\tau) \vdash 1 = 2+2\tau \in S$, so $e_1 \circ_S e_2 = \frac{1}{2}e_3$. Further

$$\begin{aligned} 1 \vdash (2+2\tau) &= \tau^2 - 2\tau - 2\tau^2 = -1 - 3\tau, \\ (2+2\tau) \vdash 1 &= 2\tau^2 + 2\tau^3 - \tau = 4 + 5\tau, \\ (1+\tau) \vdash (2+2\tau) &= \tau^2 + \tau^3 - 2\tau - 2\tau^2 = -\tau, \\ (2+2\tau) \vdash (1+\tau) &= 2\tau^2 + 2\tau^3 - \tau - \tau^2 = 3 + 4\tau, \end{aligned}$$

and none of these lies in S, so $e_1 \circ_S e_3 = e_2 \circ_S e_3 = 0$. For $x = e_1 + e_2$ this gives

$$\begin{aligned} x \circ_S x &= e_1 + e_2 + 2 \cdot \tfrac{1}{2}e_3 = e_1 + e_2 + e_3, \\ (x \circ_S x) \circ_S (x \circ_S x) &= e_1 + e_2 + e_3 + 2 \cdot \tfrac{1}{2}e_3 = e_1 + e_2 + 2e_3, \\ (x \circ_S x) \circ_S x &= e_1 + \tfrac{1}{2}e_3 + \tfrac{1}{2}e_3 + e_2 = e_1 + e_2 + e_3, \\ \big((x \circ_S x) \circ_S x\big) \circ_S x &= e_1 + e_2 + e_3, \end{aligned}$$

the last line because $(x \circ_S x) \circ_S x = x \circ_S x$. The difference of the second and fourth lines is e_3.

Large truncations. Let $x \in \mathfrak{J}(\Xi)$ have at least two points in its support. For $u, v \in \mathfrak{J}(\Xi)$ write $P(u,v) = \{s \vdash t,\ t \vdash s : s \in \operatorname{supp} u,\ t \in \operatorname{supp} v\}$, a finite set containing $\operatorname{supp}(u \circ v)$ by (5), and let

$$T = \operatorname{supp} x \cup P(x,x) \cup P(x^2,x) \cup P(x^2,x^2) \cup P(x^3,x),$$

the products being taken in $\mathfrak{J}(\Xi)$. Let $S \supseteq T$ be finite. In each of the four products $x \circ x$, $x^2 \circ x$, $x^2 \circ x^2$, $x^3 \circ x$, every quasiaddition that occurs in (5) lands in $T \subseteq S$, so under either reading of the truncation no term is removed, and the products computed in $\mathfrak{J}_S$ coincide with those computed in $\mathfrak{J}(\Xi)$. By Theorem A they violate $x^2x^2 = x^3x$. When $\Xi = \mathcal{F}(\Omega)$ is a Fibonacci chain, T is a finite subset of the discrete set $\Xi \subset \mathbb{R}$, so every set of consecutive points of the chain that reaches from $\min T$ to $\max T$ contains T. □

Theorem B removes the finite-dimensional algebras that [7, Section 6] proposes to feed into Svinolupov's classification [11]: that classification concerns structure constants of Jordan algebras, and these truncations are not Jordan algebras.

6. Ideals, units and subalgebras: proof of Theorem C

Proof of Theorem C(a). In [7, Section 5] the generators of $\mathfrak{J}(\mathcal{F}_{1,0})$ are indexed by $n \in \mathbb{Z}$ through the n-th point of the chain, so that their L_0, L_1, L_3 are our L_1, $L_{1+\tau}$, $L_{2+3\tau}$, and $\mathfrak{k} = \{L_1 \circ u : u \in \mathfrak{J}(\mathcal{F}_{1,0})\}$. By (5), every element of $\mathfrak{k}$ is a combination of generators $L_{1 \vdash y}$ and $L_{y \vdash 1}$ with $y \in \mathcal{F}_{1,0}$. The generator $L_{1+\tau}$ is not among them: $1 \vdash y = 1+\tau$ means $\tau^2 - \tau y = \tau^2$, so $y = 0$; and $y \vdash 1 = 1+\tau$ means $\tau^2 y = 1+2\tau = \tau^3$, so $y = \tau$; neither 0 nor τ belongs to $\mathcal{F}_{1,0}$. So the coefficient of $L_{1+\tau}$ vanishes on $\mathfrak{k}$. On the other hand, by Example 4.2, $L_1 \circ L_{1+\tau} = \frac{1}{2}(L_{-\tau} + L_{2+2\tau}) \in \mathfrak{k}$, and

$$\begin{aligned} (-\tau) \vdash (2+3\tau) &= -\tau^3 - 2\tau - 3\tau^2 = -4 - 7\tau, \\ (2+3\tau) \vdash (-\tau) &= 2\tau^2 + 3\tau^3 + \tau^2 = 6 + 9\tau, \\ (2+2\tau) \vdash (2+3\tau) &= 2\tau^2 + 2\tau^3 - 2\tau - 3\tau^2 = 1 + \tau, \\ (2+3\tau) \vdash (2+2\tau) &= 2\tau^2 + 3\tau^3 - 2\tau - 2\tau^2 = 3 + 4\tau, \end{aligned}$$

so that

$$(L_1 \circ L_{1+\tau}) \circ L_{2+3\tau} = \tfrac{1}{4}\left(L_{-4-7\tau} + L_{6+9\tau} + L_{1+\tau} + L_{3+4\tau}\right),$$

whose coefficient of $L_{1+\tau}$ is $\frac{1}{4}$. Hence $\mathfrak{k} \circ \mathfrak{J}(\mathcal{F}_{1,0}) \not\subseteq \mathfrak{k}$. □

Proof of Theorem C(b). Let $\varepsilon\big(\sum a_x L_x\big) = \sum a_x$. By (2), $\varepsilon(L_x \circ L_y) = 1 = \varepsilon(L_x)\varepsilon(L_y)$, so ε is an algebra homomorphism onto $\mathbb{R}$. Its kernel is an ideal; it is proper because $\varepsilon(L_x) = 1$, and nonzero because it contains $L_a - L_b$ for distinct $a, b \in \Xi$. So $\mathfrak{J}(\Xi)$ is not simple. □

Proof of Theorem C(c). No unit. Let $I = \sum \xi_x L_x$ satisfy $I \circ L_s = L_s$ for all $s \in \Xi$. By (5), $I \circ L_s = \frac{1}{2}\sum_x \xi_x (L_{x\vdash s} + L_{s\vdash x})$, and by Lemma 2.1(c) the generator L_s occurs in $L_{x\vdash s} + L_{s\vdash x}$ only for $x = s$, where it occurs twice. So the coefficient of L_s in $I \circ L_s$ is ξ_s, and $\xi_s = 1$ for every $s \in \Xi$. Since Ξ is infinite by Lemma 2.1(d), this contradicts the finiteness of $\operatorname{supp} I$.

Squares grow. Let $u \in \mathfrak{J}(\Xi)$ have at least two points in its support, choose f as in the proof of Theorem A, and let $u_0 = u$, $u_{k+1} = u_k \circ u_k$. By Lemma 3.1 and induction, every u_k is f-sharp with $M_f(u_k) > m_f(u_k)$ and $M_f(u_{k+1}) > M_f(u_k)$. The highest point of u_k therefore has a larger value of f than every point in the supports of $u_0, \dots, u_{k-1}$, so $u_0, u_1, u_2, \dots$ are linearly independent.

Subalgebras. If $\mathbb{R}u$ is a subalgebra with $u \neq 0$, then $u_1 = u \circ u \in \mathbb{R}u$, so u cannot have two points in its support, since then u_0, u_1 would be independent; hence $u = cL_x$, and conversely $\mathbb{R}L_x$ is a subalgebra because $L_x \circ L_x = L_x$. If B is a subalgebra of dimension at least two, it contains an element with at least two points in its support: take independent $v, w \in B$; if neither has two points in its support, then $v = cL_s$ and $w = dL_t$ with $s \neq t$, and $v + w \in B$ has two. That element generates the infinite independent family $u_0, u_1, \dots$ inside B, so B is infinite-dimensional. □

Proof of Theorem C(d). A subalgebra of a Jordan algebra satisfies the Jordan identity, and $\mathfrak{J}(\Xi)$ does not, by Theorem A. The extension of [8, Section 5] is $\mathfrak{J}(\Xi) \oplus \mathbb{R}$ with a product that restricts to (2) on $\mathfrak{J}(\Xi)$, as stated there, so it is not a Jordan algebra. □

Declarations

Data availability. All results are proved in the text. A script that rechecks in exact arithmetic the finite computations of Example 4.2, Theorem B and Theorem C(a) is included with the source files of this paper (file `anc/verify_jordan.py`) and archived, with the LaTeX source, at Zenodo, doi:10.5281/zenodo.24123095.

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